STACKS, DIFFERENTIAL GRADED SHEAVES: LEMMA-SECOND-PROPERTY-DG-INJECTIVE

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 4023-4039 of sources/stacks-sdga/pinned-original-source.tex; proof span is lines 4041-4059, the complete author proof (17 body lines).

Editorial note. Navigation stubs: 1 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 39 of queue_sdga.json (pb=736, stmt 4023-4039, proof 4041-4059).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 0FSQ: sources/stacks-sdga/pinned-original-source.tex, source lines 3731–3773. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $(\mathcal{C}, \mathcal{O})$ be a ringed site. Let $(\mathcal{A}, d)$ be a sheaf of differential graded algebras on $(\mathcal{C}, \mathcal{O})$. Let $\mathcal{I}$ be a K-injective and graded injective object of Mod$(\mathcal{A}, d)$. For every solid diagram in Mod$(\mathcal{A}, d)$*

$$\begin{array}{ccc} \mathcal{M} & \xrightarrow{a} & \mathcal{I} \\ {\scriptstyle b}\downarrow & \nearrow & \\ \mathcal{M}' & & \end{array}$$

where b is a quasi-isomorphism a dotted arrow exists making the diagram commute up to homotopy.

Proof. After replacing $\mathcal{M}'$ by the direct sum of $\mathcal{M}'$ and the cone on the identity on $\mathcal{M}$ (which is acyclic) we may assume b is also injective. Then the cokernel $\mathcal{Q}$ of b is acyclic. Thus we see that

$$\mathrm{Hom}_{K(Mod(\mathcal{A},\mathrm{d}))}(\mathcal{Q}, \mathcal{I}) = \mathrm{Hom}_{K(Mod(\mathcal{A},\mathrm{d}))}(\mathcal{Q}, \mathcal{I})[1] = 0$$

as $\mathcal{I}$ is K-injective. As $\mathcal{I}$ is graded injective by Remark 0FSQ we see that

$$\mathrm{Hom}_{K(Mod(\mathcal{A},\mathrm{d}))}(\mathcal{M}', \mathcal{I}) \longrightarrow \mathrm{Hom}_{K(Mod(\mathcal{A},\mathrm{d}))}(\mathcal{M}, \mathcal{I})$$

is bijective and the proof is complete. □

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